Exact versus Kohn–Sham for a one-dimensional two-electron atom

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1 Problem statement

Goal. Build a system whose exact ground state can be computed, then use it to measure how good approximate density functional theory is. Real atoms beyond helium never allow this, which is why a one-dimensional toy world is used.

System. A nucleus of charge Z sits at $x = 0$ on a line. Two electrons move on the line. Every Coulomb interaction is softened:

$$w(x) = \frac{1}{\sqrt{x^2 + 1}}, \qquad \hat{H} = -\tfrac{1}{2}\partial_{x_1}^2 - \tfrac{1}{2}\partial_{x_2}^2 - Z\,w(x_1) - Z\,w(x_2) + w(x_1 - x_2). \tag{1}$$

Atomic units are used throughout: lengths in bohr, energies in hartree (Ha), $\hbar = m_e = e = 1$.

Phases of the project.

Phase 1. One-electron solver: uniform grid on $[-L, L]$, three-point finite-difference kinetic energy, tridiagonal eigensolver, error $O(h^2)$, Richardson extrapolation.

Phase 2. Exact two-electron solver: a sparse two-dimensional Hamiltonian on the (x_1, x_2) grid and its lowest eigenstate. Extract the density. The lowest antisymmetric state (the triplet) comes for free.

Phase 3. Kohn–Sham with exact exchange, which equals restricted Hartree–Fock for two electrons. The exact correlation energy is then $E_c = E_{\text{exact}} - E_{\text{HF}}$.

Phase 4. LDA exchange for the 1D soft-Coulomb electron gas, derived from scratch.

Phase 5. LDA correlation from a published parametrisation.

Phase 6. The exact exchange–correlation potential of the two-electron system by direct inversion.

Extension. Stretched H_2: two soft nuclei at $\pm R/2$ with nuclear repulsion $w(R)$. **Parameter studies.** Grid convergence (mandatory), SCF mixing, a scan over Z, a scan over the softening parameter, and the HOMO test $-\varepsilon_{\text{HOMO}} = I$. The problem statement advises finishing Phases 1–3 first.

What we completed. Phase 1 (including Richardson extrapolation); Phase 2 for H^-, He and H_2; Phase 3 for the same three systems; the stretched-H_2 extension (exact against restricted Hartree–Fock); and the studies of grid convergence, box size, SCF mixing and the HOMO test for Hartree–Fock.

What we did not complete. Phase 4 was derived on paper but *not implemented*, so this report contains no LDA numbers of our own. Phase 5 (LDA correlation), Phase 6 (the exact exchange–correlation potential), the scans over Z and over the softening parameter, the triplet state, and measured time and memory scaling were not done either. Section 9 sets out how they would be done.

So the project compares the exact solution with Hartree–Fock. That measures the exact correlation energy, and with it the full error of Hartree–Fock, but it does *not* yet measure the error of an approximate density functional.

1.1 Code and repository

All code is in the repository `https://github.com/PiCamel/Exact-Against-HF-1D-Soft-Coulomb-Solver`. It is organised by method, with one file per system (Table 1).

Every solver file has a `run(...)` function that returns its results as a dictionary (energy, density, grid and, for Kohn–Sham, the energy terms and convergence history), and can also be run on its own (`python KS/KS_He.py`). Running `python compare.py` reproduces everything in Section 7 in about a minute and a half.

Path	Contents
`common.py`	Shared building blocks: grid, soft-Coulomb w, external potentials, kinetic matrix, two-electron Hamiltonian, Hartree potential, one-electron solver, Richardson extrapolation and the observed order of convergence
`exact/`	Exact solvers (Phase 2): `exact_H_minus.py`, `exact_He.py`, `exact_H2.py`
`KS/`	Kohn–Sham with exact exchange = Hartree–Fock (Phase 3): `KS_H_minus.py`, `KS_He.py`, `KS_H2.py`
`compare.py`	Runs every solver, prints every number quoted in Section 7 and saves the three plots
`results/`	The plots: `energy_errors.png`, `H2_energy_curve.png`, `densities.png`
`report/`	This report (`report.tex`, `report.pdf`)
`README.md`, `PS.md`	Project description and the original problem statement

Table 1: Layout of the repository.

2 Why density functional theory exists

The exact wavefunction $\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N)$ depends on the position of *every* electron at once. If each coordinate sits on a grid of N_g points in d dimensions, then

$$\text{unknowns (exact)} = \left(N_g^d\right)^N, \qquad \text{unknowns (orbitals)} \approx N \cdot N_g^d. \tag{2}$$

In plain words

Two dice need a 6×6 table of joint probabilities (36 numbers). If the dice ignore each other, two lists of 6 numbers (12 numbers) are enough. Ten dice need $6^{10} \approx 6 \times 10^7$ numbers for the joint table, but only 60 for independent dice. The exact wavefunction is the joint table. The orbital picture is the pair of independent lists.

For two electrons with $N_g = 50$ in 3D, the exact Ψ has $50^6 \approx 1.6 \times 10^{10}$ entries (about 125 GB) while one orbital has 125,000. Density functional theory (DFT) avoids the wall by working with the density $n(x)$, a function of one position, and moving all the difficult physics into a single unknown term that is approximated [1, 2]. Our toy model keeps the exact answer available, so the cost of this trade can be measured exactly.

3 The exact solution on a grid (Phases 1 and 2)

3.1 Symmetry, spin and the density

The Hamiltonian (1) contains no spin. Its lowest-energy position function has no nodes, so it is symmetric under $x_1 \leftrightarrow x_2$. A symmetric position function must be paired with the antisymmetric spin singlet (opposite spins), which satisfies the Pauli principle. So we solve for one real, positive, symmetric $\Psi(x_1, x_2)$. The density counts either electron:

$$n(x) = 2 \int |\Psi(x, x')|^2 \, \mathrm{d}x', \qquad \int n(x) \, \mathrm{d}x = 2. \tag{3}$$

3.2 Finite difference method

Store a function only at grid points $x_i = -L + ih$. An integral becomes a sum, $\int f \, \mathrm{d}x \to h \sum_i f_i$. A second derivative becomes a difference of neighbours. Adding the Taylor expansions of $\psi(x \pm h)$ cancels the odd terms:

$$\psi''(x_i) \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{h^2} \qquad (\text{error } O(h^2)). \tag{4}$$

Then $-\frac{1}{2}\partial_x^2$ is a tridiagonal matrix T with $1/h^2$ on the diagonal and $-1/(2h^2)$ beside it, a potential is a diagonal matrix, and the Schrödinger equation becomes a matrix eigenvalue problem

$$(T + V)\boldsymbol{\psi} = E\boldsymbol{\psi}. \tag{5}$$

The ground state is the lowest eigenvector. This is guaranteed by the variational principle: for any trial vector, $\langle\psi|\hat{H}|\psi\rangle/\langle\psi|\psi\rangle \geq E_0$. In the code, T is `T_1e` and the one-electron solver is `one_electron`, both in `common.py`.

3.3 Two electrons: the Kronecker product

$\Psi(x_1, x_2)$ becomes an $N \times N$ table, flattened into a vector of N^2 entries (index $k = iN + j$). Each electron's kinetic energy acts along its own index:

$$H_{2e} = T \otimes I + I \otimes T + \mathrm{diag}(V), \qquad V_{ij} = v(x_i) + v(x_j) + w(x_i - x_j). \tag{6}$$

For $N = 201$ this matrix has 40,401 rows but only about 5 non-zeros per row, so it is stored sparse and the lowest eigenvalue is found by the Lanczos method (`eigsh`), which only needs matrix–vector products. Normalisation: an eigenvector has $\sum \psi_i^2 = 1$, but physics needs $\int \psi^2 \,\mathrm{d}x = h \sum \psi_i^2 = 1$, so $\psi = \text{vector}/\sqrt{h}$ in 1D and $\Psi = \text{vector}/h$ in 2D. In the code this Hamiltonian is `H_two_electron` in `common.py`.

3.4 Richardson extrapolation

If the error is $c\,h^2$, then $E(h) = E_0 + ch^2$ and $E(2h) = E_0 + 4ch^2$. Subtracting,

$$E_0 \approx E(h) + \frac{E(h) - E(2h)}{3}. \tag{7}$$

Check: $E(h) - E(2h) = -3ch^2$, so the correction $-ch^2$ cancels the error in $E(h)$. With three spacings $h, 2h, 4h$ the observed order of convergence is $p = \log_2\big[(E_{2h} - E_{4h})/(E_h - E_{2h})\big]$, which should be 2. Both formulas are in `common.py` (`richardson`, `convergence_order`) and are used by `compare.py`.

Insight from the project

Always check $h\sum_i n_i = 2$ and the h^2 scaling. In this project the integral of n came out as 2.0000000 for every solver, and the helium energy converged with successive differences shrinking by factors 4.07 and 4.02 (Table 5). Forgetting a factor h in a sum that stands in for an integral is the most common silent bug.

4 Density functional theory: Hohenberg–Kohn and Kohn–Sham

Hohenberg–Kohn. The ground-state density determines the external potential and therefore everything [1]. The energy is a functional of the density, $E[n] = F[n] + \int n\, v_{\mathrm{ext}}\,\mathrm{d}x \geq E_0$, minimised by the true density. The catch is that nobody knows the kinetic-energy part of $F[n]$ exactly.

Kohn–Sham. Replace the interacting electrons by non-interacting ones that have the *same density*. Their kinetic energy T_s is easy to compute from orbitals. Everything else is gathered in one unknown, the exchange–correlation energy [1, 2, 3]:

$$E[n] = T_s[n] + \int n\, v_{\mathrm{ext}}\,\mathrm{d}x + E_H[n] + E_{xc}[n], \qquad \Big[-\tfrac{1}{2}\partial_x^2 + v_{\mathrm{ext}} + v_H + v_{xc}\Big]\varphi_i = \varepsilon_i\varphi_i, \tag{8}$$

with $v_{xc} = \delta E_{xc}/\delta n$ and $n = 2\sum_{\mathrm{occ}} |\varphi_i|^2$. If E_{xc} were exact, DFT would be exact. All practical error comes from approximating it. Section 5.2 shows which E_{xc} our calculation uses.

In plain words

The Kohn–Sham electrons are stunt doubles. They are not the real actors (they do not interact), but they are dressed so that the audience (the density) cannot tell the difference. All the acting the doubles cannot do is handled by v_{xc}.

5 Hartree–Fock, self-interaction and the SCF loop (Phase 3)

5.1 One orbital for both electrons

The cheapest wavefunction puts both electrons (opposite spins) in the same position orbital, $\Psi(x_1, x_2) = \varphi(x_1)\varphi(x_2)$, with density $n = 2\varphi^2$. The Hartree potential is the electrostatic potential of the whole electron cloud:

$$v_H(x) = \int n(x')\, w(x - x')\, \mathrm{d}x', \qquad E_H = \tfrac{1}{2}\int n\, v_H\, \mathrm{d}x = \tfrac{1}{2}\iint n(x)\, n(x')\, w(x - x')\, \mathrm{d}x\, \mathrm{d}x'. \tag{9}$$

On the grid this is a matrix–vector product, $v_H = h\, W n$ with $W_{ij} = w(x_i - x_j)$. The factor h is the integration weight.

Where the $\frac{1}{2}$ in E_H comes from. The density n holds the charge of both electrons, and $\int n\, v_H\, \mathrm{d}x$ adds up the repulsion between every piece of that cloud and every other piece. The double integral visits each pair of points twice, once as (x, x') and once as (x', x), so without the $\frac{1}{2}$ every repulsion would be counted twice. The $\frac{1}{2}$ removes this double counting. It is a different $\frac{1}{2}$ from the one in Section 5.2: this one appears for any number of electrons, while that one is special to two.

5.2 Equivalence of HF and KS for this problem

The cloud n contains *both* electrons, so v_H also makes each electron repel its own charge. That self-repulsion is unphysical. For two electrons in one orbital each electron is exactly half the cloud, so exact exchange removes exactly half of the Hartree energy and potential:

$$E_x = -\tfrac{1}{2}E_H, \qquad v_x = -\tfrac{1}{2}v_H, \qquad v_{\text{eff}} = v_{\text{ext}} + \tfrac{1}{2}v_H. \tag{10}$$

This is why Hartree–Fock and Kohn–Sham give *the same* equation here.

- **Hartree–Fock** minimises the energy of the product $\varphi(x_1)\varphi(x_2)$. Its orbital equation is $\big[-\tfrac{1}{2}\partial_x^2 + v_{\text{ext}} + 2J - K\big]\varphi = \varepsilon\varphi$, where J multiplies by $\frac{1}{2}v_H$ and the exchange operator K acting on the single orbital gives back $J\varphi$. So $2J - K \to J = \frac{1}{2}v_H$.
- **Kohn–Sham** with the exact-exchange functional $E_{xc} = E_x = -\frac{1}{2}E_H$ and no correlation ($E_c = 0$) has $v_{xc} = -\frac{1}{2}v_H$, so $v_{\text{ext}} + v_H + v_{xc} = v_{\text{ext}} + \frac{1}{2}v_H$.

Both give $\big[-\tfrac{1}{2}\partial_x^2 + v_{\text{ext}} + \tfrac{1}{2}v_H\big]\varphi = \varepsilon\varphi$, so the orbital, density and energy are identical. Our `KS/` solvers are therefore Hartree–Fock solvers, and this report uses "HF", "RHF" and "KS with exact exchange" for the same calculation. For more than two electrons the two methods separate: Hartree–Fock exchange becomes a non-local operator, while Kohn–Sham needs one local potential shared by all orbitals.

5.3 The energy, term by term

For the state "both electrons in φ":

$$E = \underbrace{2\langle\varphi| -\tfrac{1}{2}\partial_x^2|\varphi\rangle}_{T_s} + \underbrace{\int n\, v_{\text{ext}}\, \mathrm{d}x}_{E_{\text{ext}}} + \underbrace{\tfrac{1}{2}\int n\, v_H\, \mathrm{d}x}_{E_H} + \underbrace{\big(-\tfrac{1}{2}E_H\big)}_{E_x}. \tag{11}$$

T_s is the cost of confinement (always positive), E_{ext} is the attraction to the nucleus, E_H the cloud repelling itself, and E_x removes the self-repulsion. The correlation energy is what a product state cannot capture: electrons avoid each other, so the true Ψ is not a product. It comes for free here:

$$E_c = E_{\text{exact}} - E_{\text{HF}} \leq 0. \tag{12}$$

The nuclear repulsion E_{nn}. For H_2 both of our solvers add $E_{nn} = w(R)$: `exact_H2.py` returns `E[0] + E_nn` (and keeps the electronic energy separately as `E_elec`), and `KS_H2.py` adds `E_nn_H2` to its term-by-term sum and to the eigenvalue check. What changes with it is the *meaning* of the number:

- **Without** E_{nn} the result is the *electronic energy*: the energy of the two electrons moving in the field of two fixed nuclei. The nuclei are clamped (Born–Oppenheimer), so $w(R)$ is just a constant. Adding a constant to a Hamiltonian shifts every eigenvalue and leaves every eigenvector unchanged, so Ψ, $n(x)$ and E_c do not depend on it, as long as it is treated the same way in both methods.
- **With** E_{nn} the result is the *total energy of the whole molecule* at that geometry: electrons plus nuclei. This is the number that can be compared between different R. The electronic energy alone keeps falling as the nuclei approach (from -1.439 Ha at $R = 10$ to -2.120 Ha at $R = 1$, Table 8), so on its own it would predict no bond. The nuclear repulsion grows faster at short range, and only the sum has a minimum (the bond length) and tends to two separate hydrogen atoms as $R \to \infty$.

For the atoms there is only one nucleus, so $E_{nn} = 0$ and the electronic energy is the total energy.

Why not add the eigenvalues? Doubling the orbital equation gives $2\varepsilon = T_s + E_{\text{ext}} + E_H$, so $E = 2\varepsilon - \frac{1}{2}E_H$ (plus E_{nn} for H_2) and *not* 2ε. In general Kohn–Sham, $E = \sum_i \varepsilon_i - E_H - \int n\, v_{xc}\, \mathrm{d}x + E_{xc}$. Our code computes E both term by term and from this shortcut: two independent routes that must agree, and do, to better than 10^{-7} Ha (Table 3).

In plain words

Each ε already contains that electron's repulsion from the other one. Adding both counts the single pair twice, like counting handshakes by asking each person how many hands they shook.

5.4 The self-consistent loop

The potential depends on n and n depends on the potential, so we iterate (Figure 1). A fixed-point iteration that jumps straight to the new density can overshoot: a wider cloud weakens the repulsion at the centre, the next cloud comes out too narrow, and so on. Mixing takes only part of the step:

$$n_{\text{in}}^{(k+1)} = \alpha\, n_{\text{out}}^{(k)} + (1-\alpha)\, n_{\text{in}}^{(k)}, \qquad \|n_{\text{out}} - n_{\text{in}}\| = \sqrt{\textstyle\sum_i (n_{\text{out},i} - n_{\text{in},i})^2} < 10^{-6}. \tag{13}$$

Our code uses this plain vector norm, without the grid weight h, so the tolerance means slightly different things on different grids. That has no visible effect at the precision quoted here, but a weighted norm would be cleaner. The default is $\alpha = 0.4$, with at most 1000 iterations.

Near the answer, an error component with response eigenvalue λ is multiplied by $1 - \alpha + \alpha\lambda$ each iteration. It converges only if this lies in $(-1, 1)$, so a strongly overshooting component ($\lambda = -3$, say) needs $\alpha < 0.5$. Small α is safe but slow.

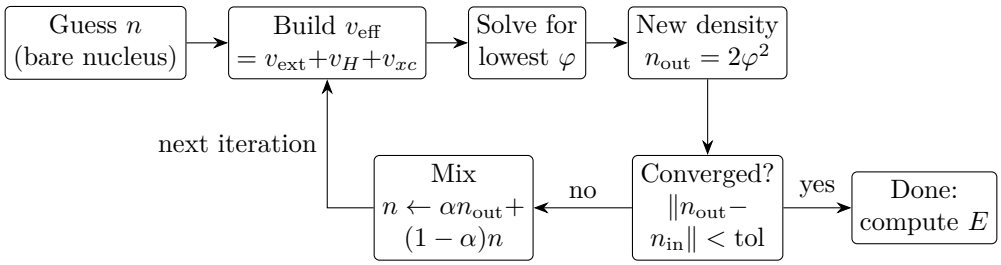


Figure 1: The self-consistent field loop. In this project $v_{xc} = -\frac{1}{2}v_H$ (exact exchange), which makes it the Hartree–Fock loop.

Insight from the project

Mixing matters only when the problem is hard. For helium, plain iteration ($\alpha = 1$) converges in 8 iterations and the fastest mixing was $\alpha = 0.9$ (6 iterations, Table 7). For stretched H_2 the same loop fails: at $R = 6$, without help, the charge sloshes from one nucleus to the other and the error is still 2.67 after 300 iterations. Forcing the density to be symmetric under $x \to -x$ at every step (the exact ground state has this symmetry) makes it converge in 22 iterations at $\alpha = 0.4$. Every H_2 result in this report uses the symmetrised loop.

6 Stretched H_2: static correlation

Place two nuclei ($Z = 1$ each) at $\pm R/2$ and add the nuclear repulsion $w(R)$ to every energy. As $R \to \infty$ the exact energy tends to two separate hydrogen atoms, $2 \times (-0.66986) = -1.33972$ Ha on our grid. Restricted Hartree–Fock keeps one orbital for both electrons. By symmetry it is $\varphi = (\chi_A + \chi_B)/\sqrt{2}$, a mix of an orbital on each nucleus, and the product is

$$\Psi_{\text{RHF}} = \tfrac{1}{2}[\chi_A(x_1)\chi_A(x_2) + \chi_B(x_1)\chi_B(x_2) + \chi_A(x_1)\chi_B(x_2) + \chi_B(x_1)\chi_A(x_2)]. \tag{14}$$

The last two terms are neutral H + H. The first two put both electrons on one atom ($H^- + H^+$). So RHF is 50% ionic at *every* distance, even when the ionic state costs a lot of energy. This is *static* (strong) correlation: the true state needs the covalent configuration alone, and one orbital cannot provide that.

In plain words

Two people in two far-apart rooms. The truth is one person per room. Restricted HF insists that each person is independently "half in each room", so half the time both are in the same room, which is wildly unlikely and costly.

Spin-unrestricted HF fixes the energy by letting each electron localise on one atom, at the cost of no longer being a proper spin singlet. We did not run it.

7 Results

Every number in this section is printed by `compare.py`, and every figure is one of its plots. The label in brackets in each caption (for example "output 1a") names the block of its terminal output that the table comes from. Unless stated otherwise the grid spacing is $h = 0.1$ and the box half-width is $L = 10$. "RHF" and "KS with exact exchange" are the same calculation (Section 5.2).

7.1 Helium and the hydrogen anion

Table 2 compares total energies. After Richardson extrapolation our solvers reproduce the problem statement's reference numbers to four decimals (output 3b): one-electron H $= -0.6698$, $\mathrm{He}^{+} = -1.4834$, exact He $= -2.2383$, RHF $= -2.2242$, $E_c = -0.0140$ and $I = 0.7548$ Ha.

System	E_{exact}	E_{RHF}	$\Delta_{\mathrm{RHF}} = -E_c$	δ_{RHF} (%)
He ($Z = 2$), $h = 0.1$	-2.23855	-2.22450	0.01405	0.63
He ($Z = 2$), extrapolated	-2.23826	-2.22421	0.01405	0.63
H^{-} ($Z = 1$), $h = 0.1$, $L = 10$	-0.73043	-0.69182	0.03861	5.29
H^{-} ($Z = 1$), $h = 0.1$, $L = 14$	-0.73079	-0.69208	0.03871	5.30

Table 2: Total ground-state energies in hartree (outputs 1a, 3b and 4). Extrapolation uses $h = 0.1$ and 0.05. $\Delta = E_{\mathrm{approx}} - E_{\mathrm{exact}}$ and $\delta = 100|\Delta|/|E_{\mathrm{exact}}|$. For Hartree–Fock, Δ is exactly minus the correlation energy.

System	T_s	E_{ext}	E_H	E_x	E_{nn}	E (terms)	E (2ε route)	iter.
H^{-}	0.1131	-1.3920	1.1743	-0.5871	0	-0.69182	-0.69182	23
He	0.2905	-3.2389	1.4477	-0.7239	0	-2.22450	-2.22450	23
H_2 ($R = 3.5$)	0.0965	-2.1652	1.0595	-0.5298	0.2747	-1.26423	-1.26423	19

Table 3: The Hartree–Fock (= KS exact exchange) energy split into its terms, in hartree (output 1b). The two routes to E differ by 5×10^{-8}, 6×10^{-8} and 1×10^{-8} Ha.

System	$I = E(\text{one electron}) - E(\text{two electrons})$	$-\varepsilon_{\mathrm{HOMO}}$ (RHF)	Δ	δ (%)
He	0.7549	0.7503	-0.0046	0.60
H^{-}	0.0606	0.0523	-0.0082	13.58

Table 4: The HOMO test at $h = 0.1$, $L = 10$ (output 1c). In exact Kohn–Sham $-\varepsilon_{\mathrm{HOMO}} = I$; for Hartree–Fock (Koopmans' theorem) it is only an estimate.

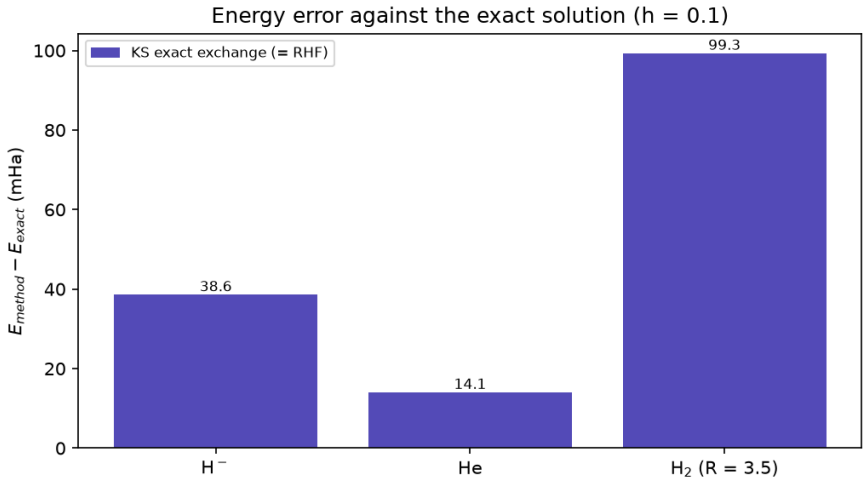


Figure 2: Energy error of Hartree–Fock against the exact solution, $E_{\mathrm{RHF}} - E_{\mathrm{exact}} = -E_c$, in millihartree. H_2 is at $R = 3.5$.

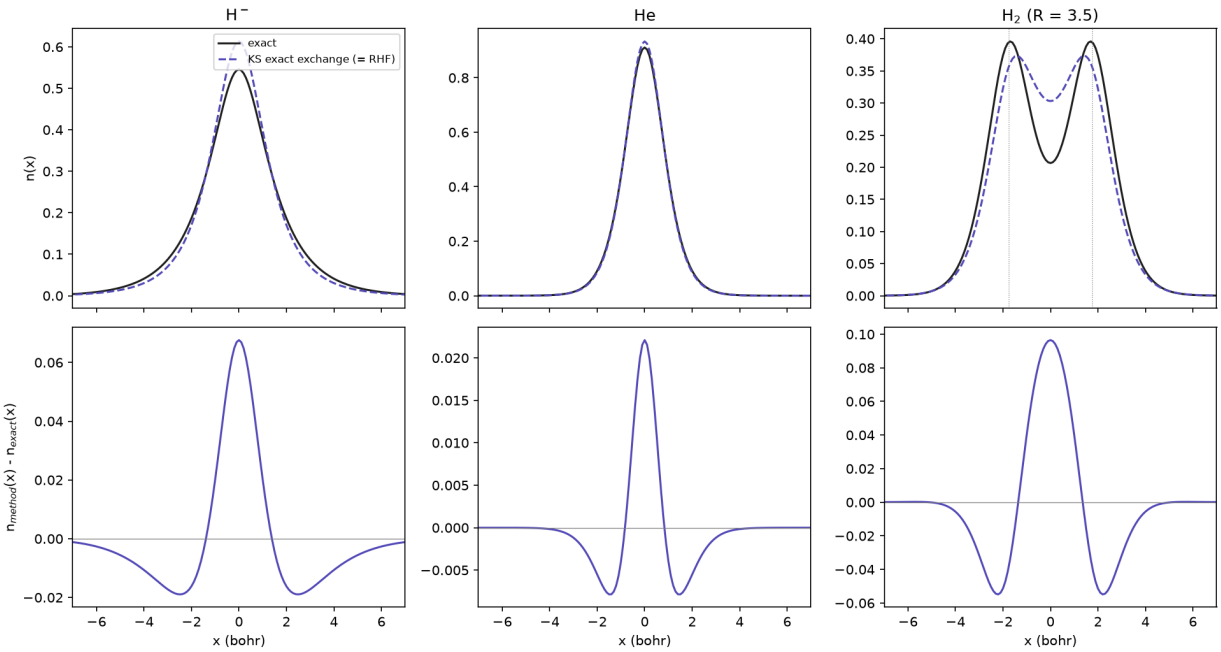


Figure 3: Top: ground-state densities, exact (solid) and Hartree–Fock (dashed). Bottom: the difference $n_{\mathrm{RHF}} - n_{\mathrm{exact}}$. Dotted lines mark the nuclei of H_2.

Insight from the project

Correlation is small in helium and large in the anion. The whole correlation energy of He is only 0.014 Ha (0.63% of the total), so Hartree–Fock is already very good there. In H^- it is 0.039 Ha, almost three times larger: the loosely bound electrons have room to avoid each other, and a product wavefunction cannot describe that avoidance. The densities in Figure 3 tell the same story (output 1d): for He the peaks nearly agree (0.933 against 0.911), while the Hartree–Fock density of H^- is too compact (peak 0.613 against 0.545).

Insight from the project

The HOMO test. The exact $-\varepsilon_{\mathrm{HOMO}}$ equals the ionisation energy. Hartree–Fock gets helium to 0.6% (0.7503 against 0.7549 Ha), because $-\frac{1}{2}v_H$ removes the self-interaction exactly and keeps the correct $-w(x)$ tail of the potential. For H^- the estimate is off by 14%: when correlation matters, Koopmans' theorem is a rough guide.

Insight from the project

Weakly bound systems are limited by the box. H^- is bound (exact $E = -0.7304$ Ha below the neutral atom's -0.6699 Ha, an electron affinity of about 0.06 Ha), but it is so diffuse that its exact energy still moves by 3.6×10^{-4} Ha between $L = 10$ and $L = 14$ (Table 6). Helium is converged to 10^{-6} Ha by $L = 8$.

7.2 Grid and box convergence

h (bohr)	E_{exact}(He)	ratio	E_{RHF}(He)	ratio
0.4	−2.243017	–	−2.228938	–
0.2	−2.239430	–	−2.225374	–
0.1	−2.238550	4.07	−2.224500	4.08
0.05	−2.238331	4.02	−2.224282	4.02
extrapolated	−2.238258	–	−2.224209	–

Table 5: Grid-spacing convergence of the helium energy at $L = 10$, in hartree (output 3a). The ratio of successive differences is 4 when the error is $\propto h^2$. Extrapolation uses $h = 0.1$ and 0.05.

The observed order of convergence, from `convergence_order`, is $p = 2.03$ and 2.01 for both solvers, as expected for a three-point second derivative.

L (bohr)	E_{exact}(He)	$E_{\mathrm{exact}}(\mathrm{H}^-)$	$E_{\mathrm{RHF}}(\mathrm{H}^-)$	$-\varepsilon_{\mathrm{HOMO}}(\mathrm{H}^-)$
6	−2.238506	−0.723313	−0.686460	0.0394
8	−2.238549	−0.729191	−0.690923	0.0492
10	−2.238550	−0.730432	−0.691823	0.0523
12	−2.238550	−0.730721	−0.692031	0.0534
14	−2.238550	−0.730790	−0.692082	0.0538

Table 6: Box-size convergence at $h = 0.1$, energies in hartree (output 4). Helium is converged by $L = 8$; H^- is still drifting at $L = 14$.

7.3 SCF mixing

α	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
He	110	52	33	23	17	13	10	7	6	8
H_2 ($R = 3.5$)	91	43	27	19	14	11	8	8	12	17

Table 7: SCF iterations to reach $\|n_{\mathrm{out}} - n_{\mathrm{in}}\| < 10^{-6}$ against the mixing parameter α (output 5a).

Helium converges for every α, fastest just below 1. For H_2 at $R = 3.5$ the optimum is $\alpha \approx 0.7$–0.8, and plain iteration ($\alpha = 1$) is already twice as slow: the extra overshoot between the two nuclei is the first sign of the sloshing that stops stretched H_2 converging at all unless the density is symmetrised (output 5b).

7.4 Stretched H_2

R	E_{exact}	E_{RHF}	$\Delta_{\text{RHF}} = -E_c$	δ_{RHF} (%)	$E_{nn} = w(R)$	exact electronic
1.0	−1.41320	−1.39452	0.01868	1.32	0.70711	−2.12030
1.5	−1.45172	−1.42639	0.02533	1.74	0.55470	−2.00642
2.0	−1.44200	−1.40591	0.03609	2.50	0.44721	−1.88921
2.5	−1.41356	−1.36158	0.05198	3.68	0.37139	−1.78495
3.0	−1.38483	−1.31146	0.07338	5.30	0.31623	−1.70106
4.0	−1.35067	−1.22322	0.12745	9.44	0.24254	−1.59321
5.0	−1.34091	−1.16115	0.17976	13.41	0.19612	−1.53703
6.0	−1.33930	−1.12054	0.21876	16.33	0.16440	−1.50370
8.0	−1.33946	−1.07697	0.26249	19.60	0.12403	−1.46350
10.0	−1.33963	−1.05706	0.28257	21.09	0.09950	−1.43913

Table 8: Stretched 1D H_2 at $h = 0.1$, with box half-width $L = \max(10, R/2 + 9)$, energies in hartree (output 2a). E_{exact} and E_{RHF} are total energies (electronic energy plus E_{nn}); the last column is the exact electronic energy alone. Mean absolute error of RHF over the 10 distances: 0.1276 Ha.

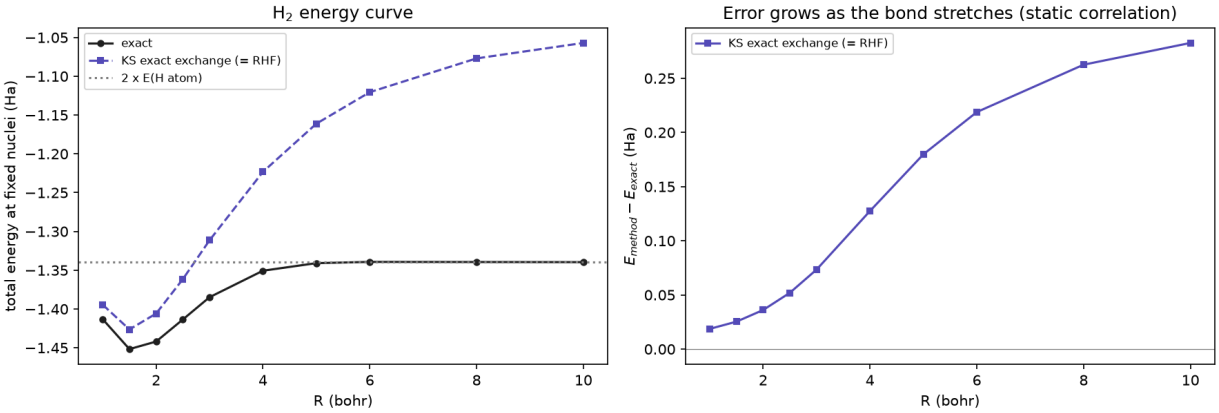


Figure 4: Left: the H_2 total-energy curve. The exact curve has its minimum near $R \approx 1.5$–2 and dissociates to two H atoms (−1.33972 Ha, grey line); RHF stays far too high. Right: the RHF error, which equals $|E_c|$ and grows to 0.28 Ha.

Insight from the project

Stretching the bond is a qualitative failure, not a small error. The RHF error rises from 0.019 Ha at $R = 1$ to 0.283 Ha at $R = 10$ and is still rising. The exact wavefunction shows why (output 2b): at $R = 3.5$ the probability that both electrons sit on the same side of the molecule is 0.145 in the exact Ψ and 0.485 in the RHF product, close to the 50% that Section 6 predicts. The lesson: say "approximate methods fail", not "the theory fails".

7.5 Agreement with the problem statement's reference table

The exact column of our stretched-H_2 table matches the problem statement's reference at $R = 1, 2, 4, 6, 10$ to within about 2×10^{-4} Ha, the size expected from its coarser grid ($h = 0.15$). The RHF column matches at $R = 1, 2$ but not beyond (Table 9).

R	RHF (reference table)	RHF (this work)
4	−1.1904	−1.2232
6	−1.0693	−1.1205
10	−1.0224	−1.0571

Table 9: Where the stretched-H_2 Hartree–Fock numbers differ from the table that came with the problem statement. All values in Ha; "this work" is output 2a.

Restricted Hartree–Fock is variational, so a lower converged energy means a better restricted state. We reproduced the likely cause ourselves: without the symmetrised density the loop does not converge for stretched H_2, because the charge sloshes between the nuclei. The reference RHF values for $R \geq 4$ were most likely taken from such an unconverged loop. We cannot check how that table was produced, so this remains our best explanation rather than a proof.

Insight from the project

Verify, then trust. Checking every energy by two routes, and checking the variational ordering $E_{\mathrm{RHF}} \geq E_{\mathrm{exact}}$, caught a disagreement with a reference table. The same discipline found the bugs of our first notebook: two different softening lengths in one Hamiltonian (0.2 for the electron repulsion of H_2, 0.001 in one H^- run), a grid spacing hard-coded separately from the grid, the Kohn–Sham loop borrowing the exact solver's grid, and a missing nuclear repulsion. Use one softening parameter for every interaction and derive h from the grid.

8 Error analysis

8.1 Definitions

Let E_{exact} be the exact value on the same grid and E_{approx} the value from a method. The signed error and the relative error are

$$\Delta = E_{\mathrm{approx}} - E_{\mathrm{exact}}, \qquad \delta = \frac{|\Delta|}{|E_{\mathrm{exact}}|} \times 100\%. \tag{15}$$

The sign matters. Hartree–Fock is variational, so $\Delta_{\mathrm{RHF}} \geq 0$ must hold for every system on the same grid. All our values satisfy this, which is a useful sanity check.

Error of the correlation energy. By definition $E_c = E_{\mathrm{exact}} - E_{\mathrm{HF}}$, so

$$\Delta_{\mathrm{RHF}} = E_{\mathrm{HF}} - E_{\mathrm{exact}} = -E_c = |E_c|. \tag{16}$$

The RHF error *is* the correlation energy.

Error of an orbital energy. Because $-\varepsilon_{\mathrm{HOMO}}$ should equal $I = E(\mathrm{ion}) - E(\mathrm{atom})$, we use $\Delta = (-\varepsilon_{\mathrm{HOMO}}) - I$.

Mean absolute error over several points. For n bond lengths,

$$\mathrm{MAE} = \frac{1}{n} \sum_{i=1}^{n} |\Delta_i|. \tag{17}$$

Over the 10 distances of Table 8, $\mathrm{MAE}_{\mathrm{RHF}} = 0.1276\,\mathrm{Ha}$.

8.2 Numerical (discretisation) uncertainty

Richardson extrapolation provides its own error estimate (output 3a). The un-extrapolated error of $E(h)$ is about $|E(h) - E(2h)|/3$, which is 0.29 mHa for helium at $h = 0.1$. A second estimate of the remaining error is the difference between two extrapolations, from $(0.2, 0.1)$ and from $(0.1, 0.05)$. For helium these are -2.238256 and -2.238258 Ha, so the extrapolated energy is good to about 10^{-6} Ha. The H_2 energies are not extrapolated; at $h = 0.1$ their grid error should be a few 10^{-4} Ha, similar to helium.

Propagation. If E_{exact} and E_{HF} each carry an independent uncertainty σ, then

$$\sigma_{E_c} = \sqrt{\sigma_{\text{exact}}^2 + \sigma_{\text{HF}}^2} \quad \text{(independent)}, \qquad \sigma_{E_c} \le \sigma_{\text{exact}} + \sigma_{\text{HF}} \quad \text{(worst case)}. \tag{18}$$

Because both solvers use the same grid, much of the h^2 error cancels in the difference, so E_c is more accurate than either energy alone: it is -0.01405 Ha both at $h = 0.1$ and after extrapolation (output 3b). All quoted errors Δ are far larger than these numerical uncertainties, except for the box-limited H^-.

8.3 Summary of results and errors

Quantity	Method	Exact	Value	Δ	δ (%)
He total energy (extrapolated)	RHF	-2.2383	-2.2242	0.0140	0.63
He $-\varepsilon_{\text{HOMO}}$ vs I	RHF	0.7549	0.7503	-0.0046	0.60
H^- total energy ($L = 10$)	RHF	-0.7304	-0.6918	0.0386	5.29
H^- $-\varepsilon_{\text{HOMO}}$ vs I ($L = 10$)	RHF	0.0606	0.0523	-0.0082	13.58
H_2 total energy, $R = 2$	RHF	-1.4420	-1.4059	0.0361	2.50
H_2 total energy, $R = 3.5$	RHF	-1.3636	-1.2642	0.0993	7.28
H_2 total energy, $R = 10$	RHF	-1.3396	-1.0571	0.2826	21.09

Table 10: Summary of the comparison with the exact solution (all values in Ha, outputs 1a, 1c, 2a and 3b). Δ = value − exact, $\delta = 100|\Delta|/|\text{exact}|$.

Insight from the project

Reading the table. Hartree–Fock is excellent for the tightly bound atom (0.63% total-energy error, 0.6% on I) because it removes self-interaction exactly and only misses correlation. It is noticeably worse for the diffuse anion (5.3%), and it is not reliable once the bond is stretched, where the error reaches 21% of the total energy. Every one of these errors is pure correlation, measured exactly, which is what makes this toy model a useful benchmark for any functional that comes next.

9 Future direction

None of this section was implemented; it is the plan for the phases we did not reach.

LDA exchange (Phase 4). The local density approximation treats every small slice of the system as a piece of uniform electron gas at the local density [2, 3]. A uniform 1D gas has $k_F = \pi n/2$ and density matrix $\gamma(x) = (k_F/\pi)\sin(k_F x)/(k_F x)$ per spin. Putting γ^2 into the exchange integral and substituting $t = k_F x$ gives

$$n\, e_x(n) = -\int_{-\infty}^{\infty} w(x)\,\gamma(x)^2\,\mathrm{d}x = -\frac{n^2}{2} J\!\left(\frac{\pi n}{2}\right), \qquad J(k) = \int_0^{\infty} \frac{\sin^2 t}{t^2\sqrt{t^2 + k^2}}\,\mathrm{d}t, \tag{19}$$

with $E_x^{\mathrm{LDA}} = \int n\, e_x(n)\, \mathrm{d}x$ and $v_x(n) = \mathrm{d}[n\, e_x(n)]/\,\mathrm{d}n$. At high density $J \approx \pi/(2k_F)$, so $e_x \to -\frac{1}{2}$, which is the check the problem statement asks for. J has no closed form, so it would be tabulated once and splined in $\ln n$. In our SCF loop only one line would change: $v_{\mathrm{eff}} = v_{\mathrm{ext}} + v_H + v_x(n)$ instead of $v_{\mathrm{ext}} + \frac{1}{2}v_H$. The problem statement's reference calculation gives -2.158 Ha for 1D helium with LDA exchange, and an exchange energy of -0.654 Ha against the exact-exchange -0.724 Ha. If that holds, LDA misses about 10% of the exchange, several times the whole correlation energy we measured (-0.014 Ha).

In plain words

LDA is like valuing hilly farmland by treating every small patch as flat land at that patch's altitude. It is fine where the land changes slowly and wrong near cliffs. Atoms are full of cliffs, because the density drops quickly.

Exact v_{xc} by inversion (Phase 6). For two electrons the Kohn–Sham orbital is exactly $\varphi = \sqrt{n/2}$, so our exact density gives the exact potential with no iteration: $v_{xc} = \varepsilon + \varphi''/(2\varphi) - v_{\mathrm{ext}} - v_H$, with $\varepsilon = -I$. Its tail should follow $-w(x)$, while LDA's dies exponentially, and for stretched H_2 it should grow a peak at the bond midpoint that no local functional can reproduce.

Also left to do. LDA correlation (Phase 5); the Z scan (our one-nucleus solvers already take `Z`); the softening scan; unrestricted Hartree–Fock for stretched H_2; and timing the solvers against grid size to measure the scaling of Section 2.

10 Conclusions

What we achieved.

1. The exact two-electron solver works for H^-, He and H_2, reproduces every reference value of the problem statement for the atoms, and is a trustworthy yardstick (about 10^{-6} Ha for extrapolated He).
2. Hartree–Fock (= Kohn–Sham with exact exchange) is implemented for the same systems. Two independent energy routes agree, and the variational ordering holds everywhere.
3. The exact correlation energy follows directly: -0.014 Ha for He, -0.039 Ha for H^-, and up to -0.28 Ha for stretched H_2, where restricted Hartree–Fock fails qualitatively because of static correlation.
4. The numerical side is under control and reproducible from `compare.py`: h^2 convergence with Richardson extrapolation, box size, SCF mixing, and the charge-sloshing failure of stretched H_2 (and its fix).

What we could not do. We did not implement LDA (Phase 4), LDA correlation (Phase 5) or the exact exchange–correlation potential (Phase 6). So the project's central question, *where approximate DFT fails*, is answered only for the special functional "exact exchange, no correlation". Section 9 describes how the remaining phases would be done.

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